

# **EMPLOYEE ATTRITION PREDICTION SYSTEM**

A Machine Learning Based Approach to Predict Employee Attrition  
Using the IBM HR Analytics Dataset

A Capstone Project Report Submitted in Partial Fulfilment of the Requirements for the  
**AIML Summer Internship 2026, IIHMF, MNNIT Allahabad, Prayagraj**

Prepared By:  
**Samyak Jain**

B.Tech. – Computer Science & Engineering  
United College of Engineering and Research, Prayagraj  
Dr. A.P.J. Abdul Kalam Technical University, Lucknow  
Academic Year: 2025–2026

## CERTIFICATE

This is to certify that the project report entitled “Employee Attrition Prediction System” has been carried out by Samyak Jain under my supervision, in partial fulfilment of the requirements of the AIML Summer Internship 2026 conducted at [Motilal Nehru National Institute of Technology](#), Allahabad and submitted as part of the Innovation and Incubation Hub MNNIT Foundation (IIHMF), MNNIT Allahabad

The work embodied in this report has not been submitted elsewhere for the award of any other degree or diploma.

Arpit Dubey

Mentor, AIML Summer Internship 2026

Innovation and Incubation Hub MNNIT Foundation (IIHMF), MNNIT Allahabad

Date: 13/07/2026

## DECLARATION

I, Samyak Jain, hereby declare that the project report titled “Employee Attrition Prediction System” is an original record of the work carried out by me during the AIML Summer Internship 2026. All the data, notebooks, and results presented in this report were produced through my own implementation and analysis, and any external resource, library, or reference used has been duly acknowledged.

I further declare that this report has not been copied from any other source and has not been submitted for the award of any other degree, diploma, or certificate.

Samyak Jain

Date: 13/07/2026

## ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who supported and guided me throughout the completion of this project, **Employee Attrition Prediction System**, undertaken as part of the **AIML Summer Internship Program 2026** by **IIHMF, MNNIT Allahabad**.

First and foremost, I would like to thank my internship mentors and instructors for providing me with the opportunity to work on this practical machine learning project. Their guidance, valuable suggestions, and continuous encouragement helped me understand the complete machine learning lifecycle, from data preprocessing to model deployment.

I am also grateful to the faculty members and coordinators of my institution for their support and motivation throughout the internship. Their encouragement inspired me to explore new concepts and improve my technical skills.

I would like to acknowledge the developers and contributors of open-source technologies such as **Python, Pandas, NumPy, Scikit-learn, Matplotlib, Plotly, Streamlit, and GitHub**, which played a significant role in the successful implementation and deployment of this project.

Finally, I would like to thank my family and friends for their constant support, patience, and encouragement throughout this learning journey. Their motivation helped me stay focused and complete this project with confidence.

This project has been a valuable learning experience that strengthened my understanding of machine learning, data analysis, model evaluation, and real-world application development. I look forward to applying these skills in future academic and professional endeavors.

## ABSTRACT

Employee attrition, the voluntary or involuntary departure of employees from an organisation, is a persistent concern for human resource departments because of the cost, disruption, and loss of institutional knowledge it causes. This project presents a machine learning based system that predicts whether an employee is likely to leave a company, using the IBM HR Analytics Employee Attrition dataset, which contains demographic, job related, and satisfaction related attributes for 1,470 employees.

The project follows a complete machine learning lifecycle: the raw dataset was studied and cleaned, categorical attributes were label encoded, and numerical attributes were standardised using a StandardScaler. Five classification algorithms - Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbours, and Support Vector Machine - were trained and compared on a held out test set, and Logistic Regression was selected as the final model based on its test accuracy. The trained model, scaler, and encoders were saved and integrated into an interactive Streamlit web application that accepts employee details as input and returns a prediction along with the probability of the employee staying or leaving, a risk indicator, and a downloadable report.

The completed application was deployed on Streamlit Community Cloud and version controlled using GitHub. This report documents the dataset, the preprocessing and modelling methodology, the exploratory analysis, the evaluation of the trained model, and the design of the deployed application, along with the limitations observed and the scope for future improvement.

## TABLE OF CONTENTS

|                                           |    |
|-------------------------------------------|----|
| CERTIFICATE .....                         | 2  |
| DECLARATION .....                         | 3  |
| ACKNOWLEDGEMENT .....                     | 4  |
| ABSTRACT .....                            | 5  |
| TABLE OF CONTENTS .....                   | 6  |
| LIST OF FIGURES .....                     | 9  |
| CHAPTER 1: INTRODUCTION .....             | 10 |
| 1.1 Employee Attrition .....              | 10 |
| 1.2 Background .....                      | 10 |
| 1.3 Importance .....                      | 10 |
| 1.4 Business Impact.....                  | 10 |
| 1.5 Problem Statement .....               | 11 |
| 1.6 Objectives .....                      | 11 |
| 1.7 Scope .....                           | 11 |
| 1.8 Applications .....                    | 12 |
| 1.9 Chapter Summary .....                 | 12 |
| CHAPTER 2: LITERATURE REVIEW .....        | 13 |
| 2.1 Employee Attrition Prediction .....   | 13 |
| 2.2 Traditional HR Methods .....          | 13 |
| 2.3 Machine Learning in HR Analytics..... | 13 |
| 2.4 Existing Research .....               | 13 |
| 2.5 Research Gap .....                    | 14 |
| 2.6 Why Logistic Regression.....          | 14 |
| 2.7 Chapter Summary .....                 | 14 |
| CHAPTER 3: DATASET DESCRIPTION.....       | 15 |
| 3.1 IBM HR Analytics Dataset .....        | 15 |
| 3.2 Source .....                          | 15 |
| 3.3 Features.....                         | 15 |
| 3.4 Target Variable.....                  | 15 |
| 3.5 Data Dictionary .....                 | 16 |
| 3.6 Dataset Statistics .....              | 16 |
| 3.7 Advantages.....                       | 17 |
| 3.8 Limitations .....                     | 17 |
| 3.9 Chapter Summary .....                 | 17 |
| CHAPTER 4: METHODOLOGY .....              | 18 |

|                                             |    |
|---------------------------------------------|----|
| 4.1 Workflow Diagram.....                   | 18 |
| 4.2 Data Collection.....                    | 18 |
| 4.3 Data Understanding .....                | 18 |
| 4.4 Data Cleaning.....                      | 18 |
| 4.5 Encoding .....                          | 18 |
| 4.6 Scaling .....                           | 19 |
| 4.7 Feature Engineering.....                | 19 |
| 4.8 Train-Test Split .....                  | 19 |
| 4.9 Model Training.....                     | 19 |
| 4.10 Model Evaluation .....                 | 20 |
| 4.11 Deployment .....                       | 20 |
| 4.12 Chapter Summary .....                  | 20 |
| CHAPTER 5: EXPLORATORY DATA ANALYSIS .....  | 21 |
| 5.1 Univariate Analysis .....               | 21 |
| 5.1.1 Employee Attrition Distribution       |    |
| 5.1.2 Age Distribution                      |    |
| 5.1.3 Monthly Income Distribution           |    |
| 5.2 Bivariate Analysis .....                | 23 |
| 5.2.1 Age vs Attrition                      |    |
| 5.2.2 Monthly Income vs Attrition           |    |
| 5.2.3 Years at Company vs Attrition         |    |
| 5.2.4 Total Working Years vs Attrition      |    |
| 5.2.5 Distance From Home vs Attrition       |    |
| 5.2.6 Work Life Balance vs Attrition        |    |
| 5.2.7 Job Satisfaction vs Attrition         |    |
| 5.2.8 Environment Satisfaction vs Attrition |    |
| 5.3 Correlation Analysis .....              | 26 |
| 5.4 Chapter Summary .....                   | 28 |
| CHAPTER 6: FEATURE ENGINEERING.....         | 29 |
| 6.1 Label Encoding .....                    | 29 |
| 6.2 StandardScaler.....                     | 29 |
| 6.3 Feature Importance .....                | 29 |
| 6.4 Business Interpretation .....           | 30 |
| 6.5 Business Recommendations.....           | 30 |
| 6.6 Chapter Summary .....                   | 31 |
| CHAPTER 7: MODEL BUILDING.....              | 32 |
| 7.1 Logistic Regression .....               | 32 |
| 7.2 Decision Tree .....                     | 32 |
| 7.3 Random Forest .....                     | 32 |
| 7.4 K-Nearest Neighbours .....              | 32 |

|                                                |    |
|------------------------------------------------|----|
| 7.5 Support Vector Machine .....               | 32 |
| 7.6 Why Logistic Regression Was Selected ..... | 32 |
| 7.7 Chapter Summary .....                      | 33 |
| CHAPTER 8: MODEL EVALUATION .....              | 34 |
| 8.1 Accuracy .....                             | 34 |
| 8.2 Confusion Matrix .....                     | 34 |
| 8.3 Classification Report .....                | 35 |
| 8.4 Probability Prediction .....               | 35 |
| 8.5 Model Comparison .....                     | 36 |
| 8.6 Final Performance .....                    | 37 |
| 8.7 Advantages.....                            | 37 |
| 8.8 Limitations .....                          | 38 |
| 8.9 Chapter Summary .....                      | 38 |
| CHAPTER 9: STREAMLIT APPLICATION .....         | 39 |
| 9.1 Architecture.....                          | 39 |
| 9.2 User Interface .....                       | 39 |
| 9.3 Prediction Flow .....                      | 39 |
| 9.4 Deployment Process .....                   | 40 |
| 9.5 GitHub Integration .....                   | 40 |
| 9.6 Application Features.....                  | 41 |
| 9.7 Chapter Summary .....                      | 41 |
| CHAPTER 10: RESULTS AND DISCUSSION.....        | 42 |
| 10.1 Model Performance .....                   | 42 |
| 10.2 Business Insights .....                   | 42 |
| 10.3 Practical Applications .....              | 42 |
| 10.4 Overall Findings .....                    | 42 |
| CHAPTER 11: CONCLUSION .....                   | 43 |
| CHAPTER 12: FUTURE SCOPE .....                 | 44 |
| 12.1 Explainable AI (SHAP) .....               | 44 |
| 12.2 Deep Learning .....                       | 44 |
| 12.3 HR Dashboard.....                         | 44 |
| 12.4 Database Integration .....                | 44 |
| 12.5 Authentication .....                      | 44 |
| 12.6 Cloud Improvements .....                  | 44 |
| 12.7 Real-time Prediction .....                | 45 |
| REFERENCES .....                               | 46 |

## LIST OF FIGURES

Figure 4.1 – End to End Project Workflow

Figure 5.1 – Attrition Count Plot

Figure 5.2 – Age Distribution Histogram

Figure 5.3 – Monthly Income Distribution

Figure 5.4 – Age vs Attrition

Figure 5.5 – Monthly Income vs Attrition

Figure 5.6 – Years at Company vs Attrition

Figure 5.7 – Total Working Years vs Attrition

Figure 5.8 – Distance From Home vs Attrition

Figure 5.9 – Work Life Balance vs Attrition

Figure 5.10 – Job Satisfaction vs Attrition

Figure 5.11 – Environment Satisfaction vs Attrition

Figure 5.12 – Overtime vs Attrition

Figure 5.13 – Correlation Heatmap

Figure 6.1 – Top 15 Important Features (Random Forest)

Figure 8.1 – Confusion Matrix - Logistic Regression

Figure 8.2 – Accuracy Comparison Chart

Figure 8.3 – ROC Curve - Logistic Regression

Figure 9.1 – Streamlit Dashboard - Home Page

Figure 9.2 – Streamlit Dashboard - Prediction Result

Figure 9.3 – GitHub Repository Screenshot

# CHAPTER 1: INTRODUCTION

## 1.1 Employee Attrition

Employee attrition refers to the gradual reduction in a company's workforce that occurs when employees leave their jobs, whether by resignation, retirement, termination, or any other reason, and are not immediately replaced. Unlike a sudden layoff, attrition is often a slow and continuous process, which makes it harder to notice until its cumulative effect becomes visible in team performance, morale, and hiring costs. For organisations of every size, understanding why employees leave and, more importantly, predicting who is likely to leave next, has become an important part of workforce planning.

## 1.2 Background

For a long time, human resource departments relied almost entirely on exit interviews and manager feedback to understand attrition after it had already happened. This reactive approach meant that by the time a pattern was identified, the organisation had usually already lost several valuable employees. With the growth of data driven decision making across business functions, HR departments have started applying analytics and machine learning to move from a reactive to a proactive approach, using historical employee data to flag employees who show early signs of disengagement or dissatisfaction, well before they decide to resign.

## 1.3 Importance

Predicting attrition in advance gives an organisation the opportunity to intervene early. Instead of reacting once an employee has already resigned, HR teams can use model predictions to identify at-risk employees, understand what factors are contributing to their dissatisfaction, and take corrective action such as revising compensation, adjusting workload, or offering career growth opportunities. This shifts the role of HR analytics from simply recording what happened to actively influencing what happens next.

## 1.4 Business Impact

Attrition is expensive. Replacing an employee typically involves the cost of recruitment, onboarding, and training, along with the less visible cost of lost productivity while the role remains vacant or while a new hire ramps up. Frequent attrition in a team can also affect the morale of remaining employees and disrupt ongoing projects. A model that can flag high-risk

employees allows an organisation to prioritise retention efforts where they are likely to matter most, rather than applying blanket retention policies across the entire workforce.

## **1.5 Problem Statement**

Given the demographic, job related, and satisfaction related details of an employee, the problem addressed in this project is to predict whether that employee is likely to leave the organisation. This is framed as a binary classification problem, where the target variable takes the value “Yes” if the employee is likely to leave and “No” if the employee is likely to stay, based on patterns learned from historical employee records in the IBM HR Analytics Employee Attrition dataset.

## **1.6 Objectives**

The objectives of this project are as follows:

1. To study and understand the IBM HR Analytics Employee Attrition dataset, including its structure, features, and target variable.
2. To clean and preprocess the dataset, including handling categorical variables through label encoding and scaling numerical variables using a StandardScaler.
3. To perform exploratory data analysis in order to understand the relationship between individual features and employee attrition.
4. To train and compare multiple classification algorithms and select the best performing model based on test accuracy.
5. To evaluate the selected model using appropriate classification metrics.
6. To deploy the trained model as an interactive Streamlit web application that can accept employee details and return a prediction in real time.

## **1.7 Scope**

This project is limited to the IBM HR Analytics Employee Attrition dataset, which represents a single simulated organisation and may not fully capture the attrition patterns of every industry or company size. The project covers the complete lifecycle from data understanding to deployment, but does not include continuous retraining on live data, integration with a

company's actual HR database, or real-time monitoring, all of which are discussed as future scope in Chapter 12.

## **1.8 Applications**

A working attrition prediction system of this kind can be used by HR departments to screen their current workforce periodically and identify employees who may need attention. It can support decisions around retention bonuses, internal transfers, workload redistribution, and manager check-ins. Beyond HR, the same classification approach can be adapted to other domains that involve predicting a binary outcome from a mix of categorical and numerical attributes, such as customer churn prediction or loan default prediction, both of which follow a very similar analytical workflow.

## **1.9 Chapter Summary**

This chapter introduced the problem of employee attrition, explained why predicting it matters from both an operational and a financial standpoint, and defined the objectives and scope of this project. The remainder of this report documents, in the order the work was actually carried out, the dataset used, the methodology followed, the exploratory analysis performed, the models trained and compared, the evaluation of the final model, and the design of the deployed Streamlit application.

## **CHAPTER 2: LITERATURE REVIEW**

### **2.1 Employee Attrition Prediction**

Employee attrition refers to the process in which employees leave an organization due to resignation, retirement, or other reasons. A high attrition rate can affect productivity, increase recruitment costs, and disrupt business operations. Predicting employee attrition helps organizations identify employees who may be at risk of leaving and allows HR teams to take timely actions to improve employee retention. In this project, machine learning techniques are used to analyze employee-related factors and predict the likelihood of attrition based on historical data.

### **2.2 Traditional HR Methods**

Traditionally, organizations relied on methods such as employee surveys, performance reviews, exit interviews, and managerial observations to understand employee turnover. Although these approaches provide useful insights, they are mostly reactive and depend heavily on human judgment. They often fail to identify employees who are likely to leave in advance. As organizations generate larger volumes of employee data, traditional methods alone are no longer sufficient for effective workforce management.

### **2.3 Machine Learning in HR Analytics**

Machine Learning has transformed Human Resource Analytics by enabling organizations to make data-driven decisions. It can analyze large amounts of employee data, identify hidden patterns, and predict future outcomes with improved accuracy. HR departments use machine learning for tasks such as employee attrition prediction, recruitment, performance analysis, and workforce planning. These predictive capabilities help organizations improve employee satisfaction and reduce unnecessary turnover.

### **2.4 Existing Research**

Several studies have applied machine learning algorithms such as Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine for employee attrition prediction. Research has shown that factors like overtime, monthly income, job satisfaction, work-life balance, and total working years significantly influence employee turnover. These studies demonstrate that predictive analytics can support HR departments in making informed retention decisions.

## **2.5 Research Gap**

Although many predictive models have achieved good accuracy, several studies focus mainly on model performance rather than practical implementation. Many solutions are not deployed as user-friendly applications, making them difficult for HR professionals to use in real-world scenarios. This project addresses this gap by developing a simple, accurate, and deployable employee attrition prediction system using Streamlit.

## **2.6 Why Logistic Regression**

Logistic Regression was selected as the final model because it is well-suited for binary classification problems such as employee attrition prediction. It provides probability-based predictions, is computationally efficient, and is easy to interpret compared to many complex algorithms. Based on the experimental results obtained in this project, Logistic Regression delivered the best overall performance and was therefore chosen for deployment.

## **2.7 Chapter Summary**

This chapter reviewed the concept of employee attrition, traditional HR practices, and the role of machine learning in HR Analytics. It also discussed existing research, identified the research gap, and explained the selection of Logistic Regression for this project. The next chapter presents the dataset used and the methodology followed to develop the employee attrition prediction system.

## CHAPTER 3: DATASET DESCRIPTION

### 3.1 IBM HR Analytics Dataset

The dataset used in this project is the **IBM HR Analytics Employee Attrition & Performance Dataset**, which is widely used for employee attrition prediction and HR analytics research. It contains employee-related information such as demographic details, job roles, compensation, work experience, and satisfaction levels. The dataset provides valuable insights into the factors that influence employee attrition, making it suitable for developing and evaluating machine learning models.

### 3.2 Source

The dataset was obtained from **Kaggle**, one of the most popular platforms for data science and machine learning projects. It is publicly available and has been extensively used in academic studies, industry projects, and predictive analytics applications related to Human Resource Management.

### 3.3 Features

The dataset contains a combination of numerical and categorical features that describe various aspects of an employee's profile and work environment. After preprocessing, **30 input features** were used for model training. Some of the important features include **Age, Department, Job Role, Monthly Income, OverTime, Work-Life Balance, Job Satisfaction, Total Working Years, Years at Company**, and **Distance From Home**. These features help the model identify patterns associated with employee attrition.

### 3.4 Target Variable

The target variable used in this project is **Attrition**, which indicates whether an employee has left the organization. It is a binary classification variable with two possible values:

- **Yes (1):** Employee is likely to leave the organization.
- **No (0):** Employee is likely to stay with the organization.

The machine learning models were trained to predict this target variable using the available employee information.

### 3.5 Data Dictionary

The dataset consists of attributes related to employee demographics, job information, compensation, performance, and satisfaction. Each feature represents a specific aspect of an employee's professional profile. For example, **Age** represents the employee's age, **MonthlyIncome** indicates monthly salary, **JobRole** specifies the employee's designation, **OverTime** shows whether the employee works overtime, and **WorkLifeBalance** reflects the employee's work-life balance rating. Together, these variables provide meaningful information for predicting employee attrition.

| Feature           | Description                                            |
|-------------------|--------------------------------------------------------|
| Age               | Age of the employee                                    |
| Department        | Department in which the employee works                 |
| JobRole           | Employee's designation                                 |
| MonthlyIncome     | Monthly salary of the employee                         |
| OverTime          | Whether the employee works overtime                    |
| JobSatisfaction   | Employee's satisfaction level                          |
| WorkLifeBalance   | Rating of work-life balance                            |
| TotalWorkingYears | Total years of professional experience                 |
| YearsAtCompany    | Number of years spent in the current company           |
| Attrition         | Target variable indicating whether the employee leaves |

### 3.6 Dataset Statistics

The IBM HR Analytics dataset contains **1,470 employee records**. After preprocessing and removing unnecessary attributes, the final dataset consisted of **30 input features** and **1 target**

**variable.** The dataset includes both numerical and categorical data types, making it suitable for applying preprocessing techniques such as label encoding and feature scaling before training machine learning models.

### 3.7 Advantages

The dataset offers several advantages for employee attrition prediction. It represents a realistic HR scenario, contains a diverse set of employee-related attributes, and includes both numerical and categorical variables. Its well-structured format makes it suitable for data analysis, visualization, feature engineering, and machine learning model development. Additionally, its widespread use in research allows meaningful comparison of different predictive models.

### 3.8 Limitations

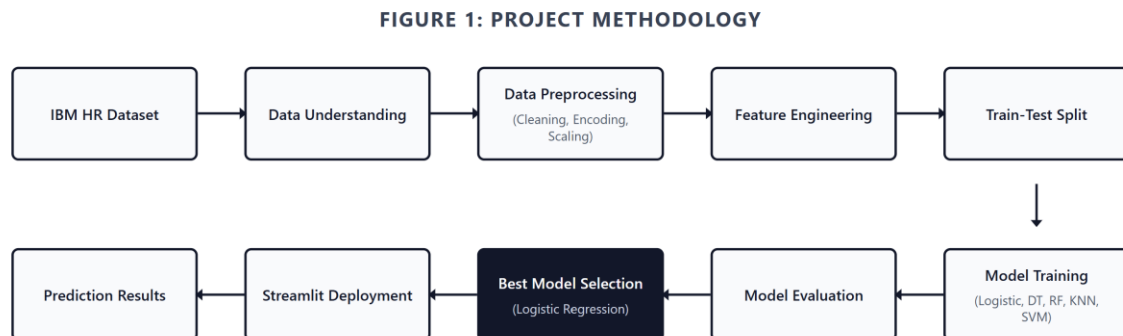
Although the dataset is comprehensive, it has certain limitations. It represents historical employee data from a single organization and may not fully capture the diversity of different industries or companies. Some real-world factors that influence employee attrition, such as organizational culture, leadership quality, and personal circumstances, are not included in the dataset. Therefore, the model's predictions should be considered as supportive insights rather than absolute decisions.

### 3.9 Chapter Summary

This chapter introduced the IBM HR Analytics dataset used in the project, along with its source, features, target variable, and overall characteristics. It also highlighted the advantages and limitations of the dataset. Understanding the dataset is an essential step before applying preprocessing techniques and building machine learning models. The next chapter explains the methodology and data preprocessing steps carried out to prepare the dataset for model development.

## CHAPTER 4: METHODOLOGY

### 4.1 Workflow Diagram



*Figure 4.1 - End-to-End Project Workflow*

### 4.2 Data Collection

The dataset used in this project is the IBM HR Analytics Employee Attrition & Performance Dataset, downloaded from Kaggle. It contains employee-related information such as demographic details, job roles, salary, experience, satisfaction levels, and attrition status. The dataset was imported into Jupyter Notebook using the Pandas library for further analysis and preprocessing.

### 4.3 Data Understanding

The dataset was explored to understand its structure, feature types, missing values, and overall quality. Basic statistical summaries, data types, and class distribution of the target variable were examined. This initial analysis provided valuable insights into the dataset before applying preprocessing techniques.

### 4.4 Data Cleaning

The dataset was carefully checked for missing values and duplicate records before model development. No missing values or duplicate rows were found; therefore, no imputation or data removal was required. The dataset was already clean and suitable for further preprocessing.

### 4.5 Encoding

Several categorical features, including **BusinessTravel**, **Department**, **EducationField**, **Gender**, **JobRole**, **MaritalStatus**, **Over18**, and **OverTime**, were converted into numerical

values using **LabelEncoder** from Scikit-learn. Separate encoders were created for each categorical feature and stored together using **joblib**, allowing the same encoding process to be applied during prediction in the deployed application.

## 4.6 Scaling

After encoding, all 34 feature columns were passed through a single **StandardScaler**, fitted on the entire feature set so that every column has zero mean and unit variance. The fitted scaler was saved as **scaler.pkl**, and the order of columns it was fitted on was separately saved as **features.pkl** to guarantee that new data is scaled and fed into the model in the exact same column order used during training.

## 4.7 Feature Engineering

Feature engineering was performed to understand the importance of different employee attributes. A Random Forest model was used to calculate feature importance scores and identify the variables that contributed most to employee attrition prediction. This analysis helped interpret the dataset but **no features were removed**, and all available features were retained for model training.

## 4.8 Train-Test Split

The processed dataset was split into training and testing sets using an 80:20 ratio with stratified sampling on the target variable, ensuring that the proportion of “Yes” and “No” labels remained consistent across both sets. A fixed **random\_state** of 42 was used to make the split reproducible.

## 4.9 Model Training

Five machine learning algorithms were trained and compared for employee attrition prediction:

- Logistic Regression
- Decision Tree
- Random Forest
- K-Nearest Neighbours (KNN)
- Support Vector Machine (SVM)

Each model was trained using the processed dataset, and their prediction performance was evaluated to identify the most suitable model.

#### 4.10 Model Evaluation

The performance of all trained models was compared using classification accuracy. In addition, evaluation metrics such as the confusion matrix, classification report, ROC-AUC score, and prediction probabilities were used to assess the performance of the selected model. Based on these results, **Logistic Regression** achieved the highest overall accuracy and was selected as the final model for deployment.

#### 4.11 Deployment

The final Logistic Regression model, along with the scaler, encoders, and feature order, was saved using **joblib**. A user-friendly web application was developed using **Streamlit**, allowing users to enter employee details and receive real-time attrition predictions. The application was deployed on **Streamlit Community Cloud**, while the complete source code and project files were maintained in a public GitHub repository.

#### 4.12 Chapter Summary

This chapter explained the complete methodology followed in developing the Employee Attrition Prediction System, starting from data collection and preprocessing to model training, evaluation, and deployment. The structured workflow ensured that the developed model was both reliable and suitable for real-world use. The next chapter presents the exploratory data analysis performed to better understand the dataset and identify important patterns related to employee attrition.

## CHAPTER 5: EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis (EDA) was performed to understand the characteristics of the dataset and identify important patterns influencing employee attrition. Different visualization techniques were used to study the distribution of variables and their relationship with employee attrition. The insights obtained during this stage helped in selecting suitable features and improving the overall understanding of the problem.

### 5.1 Univariate Analysis

#### 5.1.1 Employee Attrition Distribution

The first analysis examined the distribution of the target variable, Attrition. The visualization shows that most employees belong to the "**No Attrition**" category, while a comparatively smaller number of employees left the organization. This indicates that the dataset is moderately imbalanced, which is a common characteristic of employee attrition datasets.

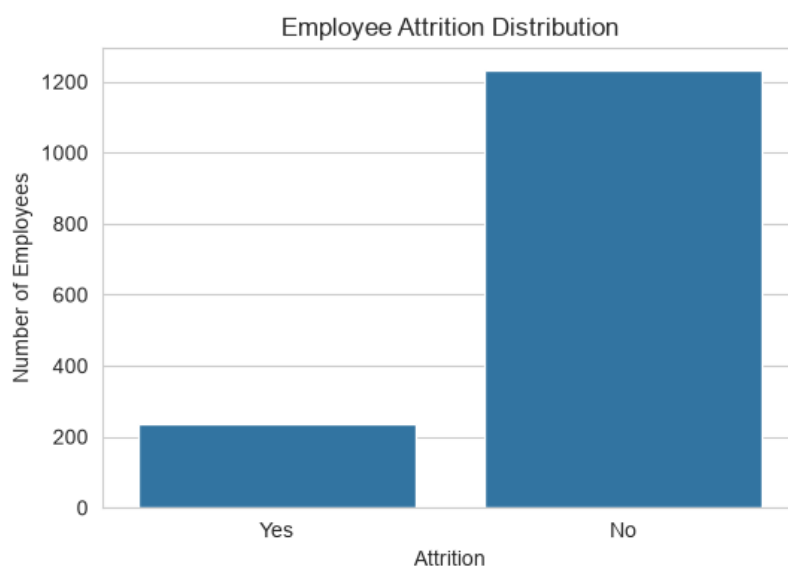


Figure 5.1: Attrition Count Plot

#### 5.1.2 Age Distribution

The age distribution reveals that most employees fall between **30 and 40 years** of age. Very few employees belong to the younger or older age groups. Understanding the age distribution helps identify whether employee age influences attrition behaviour.

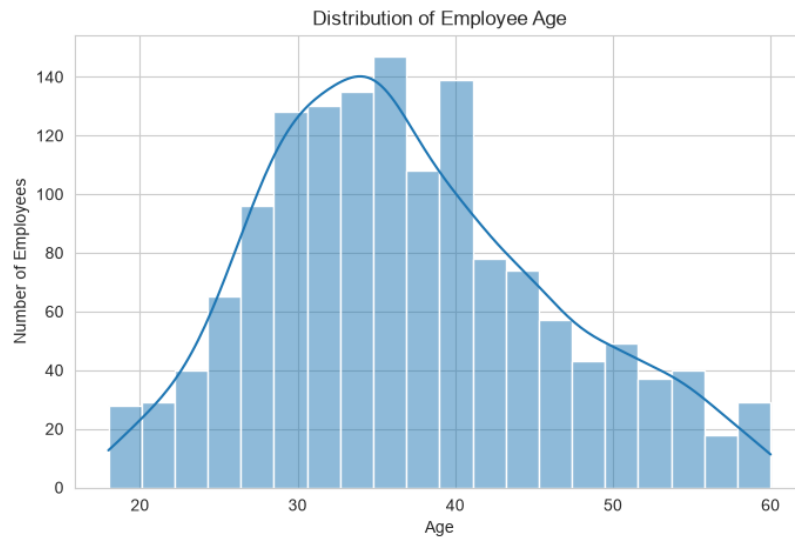


Figure 5.2: Age Distribution Histogram

### 5.1.3 Monthly Income Distribution

Monthly income shows a positively skewed distribution, with the majority of employees earning lower to moderate salaries while only a small number receive higher incomes. Salary distribution provides useful insight into employee compensation patterns within the organization.

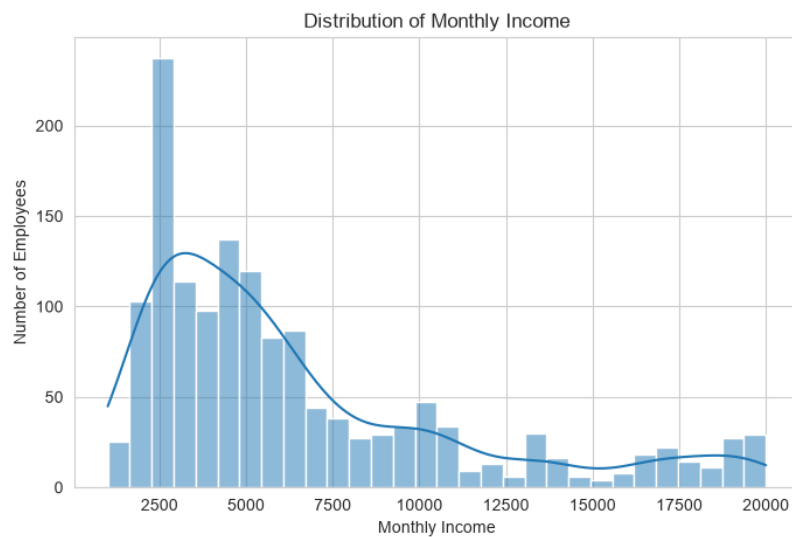


Figure 5.3: Monthly Income Distribution

## 5.2 Bivariate Analysis

### 5.2.1 Age vs Attrition

The relationship between age and attrition indicates that younger employees show a comparatively higher tendency to leave the organization. As employee age increases, the attrition rate gradually decreases. This suggests that experienced employees are generally more likely to remain with the organization, while younger professionals may seek better career opportunities.

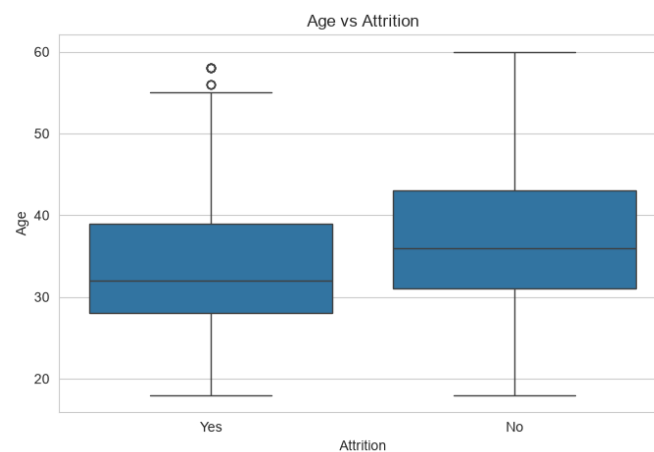


Figure 5.4: Age vs Attrition

### 5.2.2 Monthly Income vs Attrition

The analysis shows that employees with relatively lower monthly income were more likely to leave the organization. Higher salary appears to be associated with better employee retention, highlighting compensation as an important factor influencing attrition.

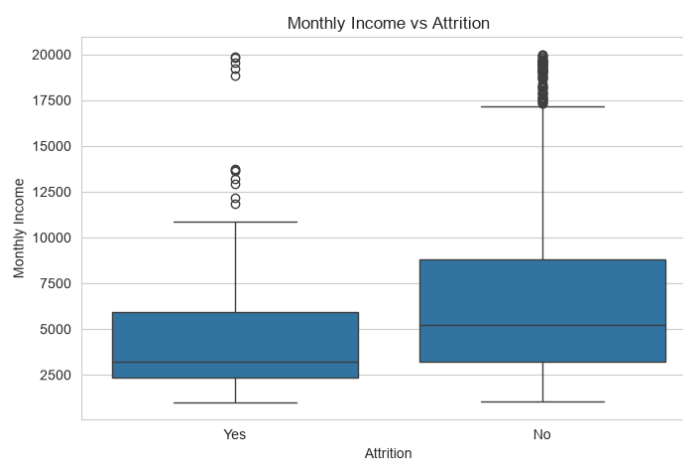


Figure 5.5: Monthly Income vs Attrition

### 5.2.3 Years at Company vs Attrition

The analysis shows that employees with fewer years at the company are more likely to leave than those who have been with the organization for a longer period. Employees who have established longer careers within the company tend to demonstrate greater loyalty and stability, resulting in lower attrition rates.

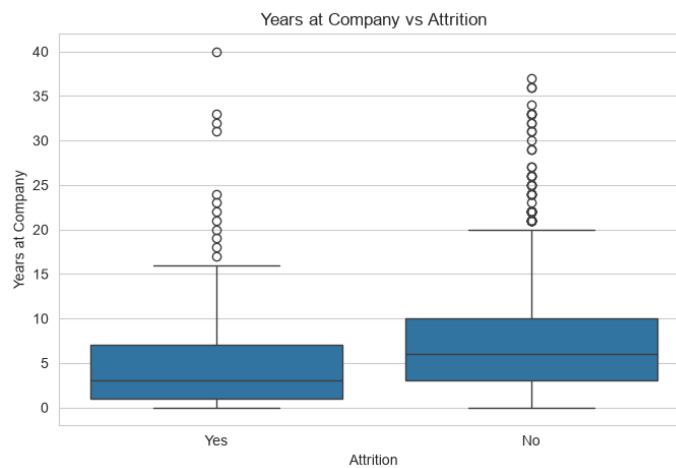


Figure 5.6: Years at Company vs Attrition

### 5.2.4 Total Working Years vs Attrition

Employees with limited overall work experience exhibit a relatively higher attrition rate compared to those with more years of professional experience. This suggests that experienced employees often have greater job stability and are less likely to switch organizations frequently.

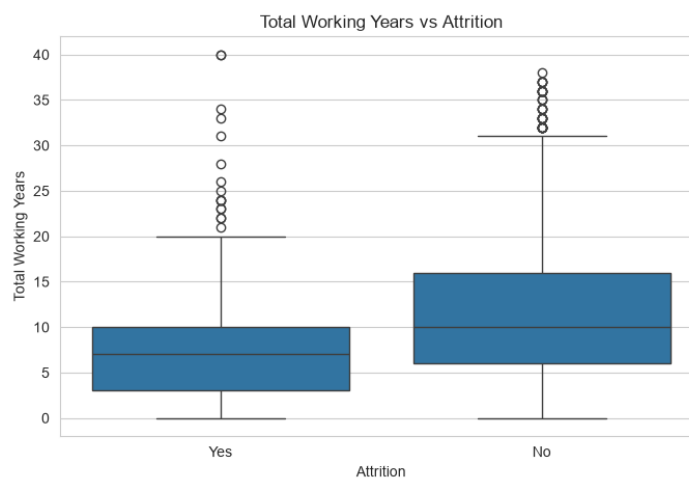


Figure 5.7: Total Working Years vs Attrition

### 5.2.5 Distance From Home vs Attrition

Employees living farther from the workplace tend to show slightly higher attrition compared to those residing closer. Although distance alone is not a deciding factor, commuting convenience may influence long-term employee retention.

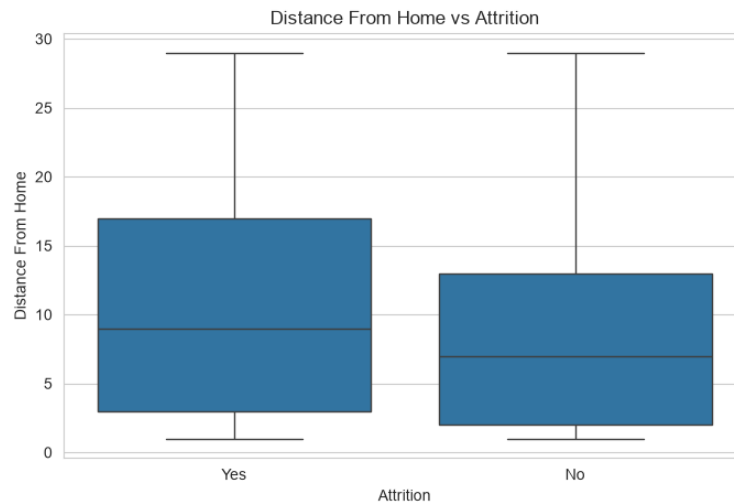


Figure 5.8: Distance From Home vs Attrition

### 5.2.6 Work-Life Balance vs Attrition

Employees reporting lower work-life balance ratings experienced higher attrition. Maintaining a healthy balance between professional and personal life appears to improve employee satisfaction and reduce the likelihood of resignation.



Figure 5.9: Work-Life Balance vs Attrition

### 5.2.7 Job Satisfaction vs Attrition

The visualization indicates that employees with lower job satisfaction are more likely to leave the organization. Higher satisfaction levels generally correspond to improved employee retention, emphasizing the importance of workplace engagement.

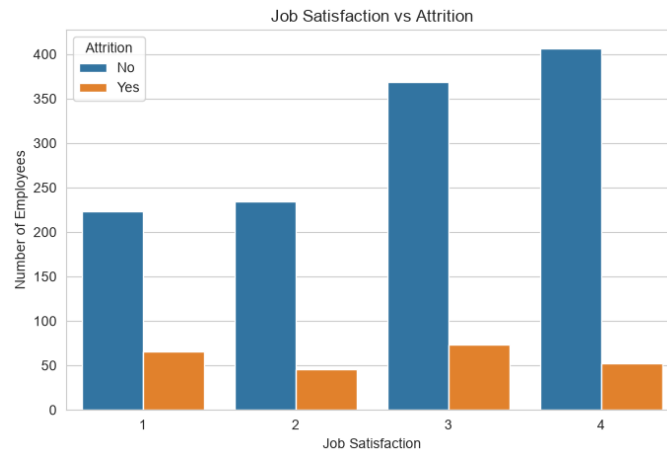


Figure 5.10: Job Satisfaction vs Attrition

### 5.2.8 Environment Satisfaction vs Attrition

Employees with lower environment satisfaction ratings show a higher tendency to leave the organization. In contrast, employees who are satisfied with their work environment generally exhibit lower attrition rates. This highlights the importance of maintaining a positive and supportive workplace culture for improving employee retention.

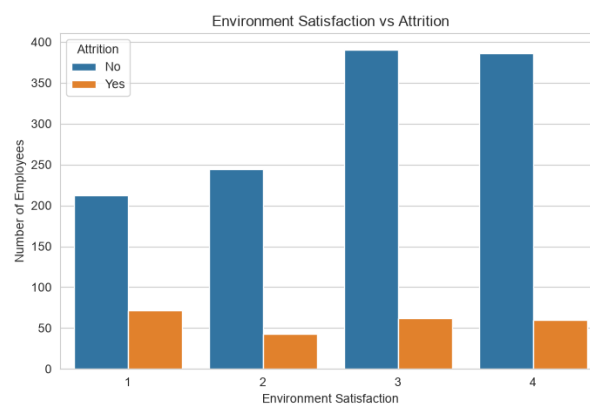


Figure 5.11: Environment Satisfaction vs Attrition

### 5.2.9 OverTime vs Attrition

Employees who frequently worked overtime showed a considerably higher attrition rate compared to those who did not. This suggests that excessive workload and work-related stress may contribute significantly to employee turnover.

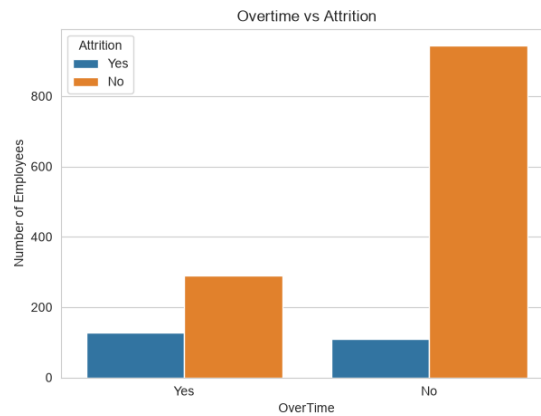


Figure 5.12: OverTime vs Attrition

## 5.3 Correlation Analysis

A correlation heatmap was generated to examine relationships among numerical variables in the dataset. Most features exhibited weak to moderate correlations, indicating that they provide unique information for model training. Some variables, such as **Monthly Income**, **Job Level**, and **Total Working Years**, showed relatively stronger positive relationships, while no severe multicollinearity was observed. This confirms that the dataset is suitable for machine learning model development.

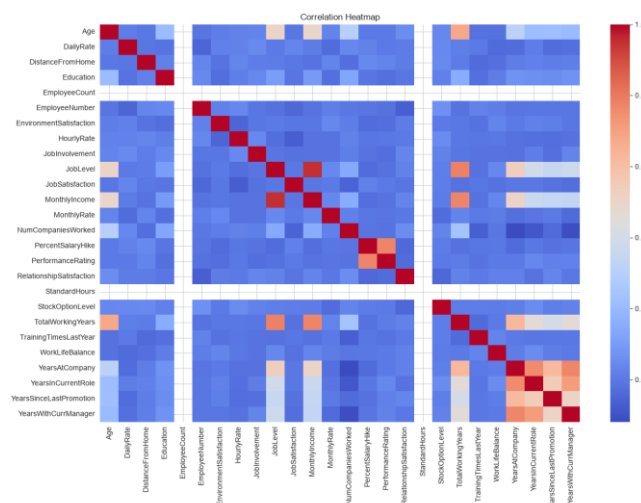


Figure 5.13: Correlation Heatmap

## 5.4 Chapter Summary

This chapter presented the exploratory analysis of the IBM HR Analytics dataset using various statistical visualizations. The findings highlighted important factors such as overtime, job satisfaction, work-life balance, monthly income, and job role that influence employee attrition. These insights provided a strong foundation for the machine learning models developed in the subsequent chapters.

## CHAPTER 6: FEATURE ENGINEERING

Feature engineering plays a crucial role in improving the performance of machine learning models by transforming raw data into a suitable format for analysis. In this project, categorical variables were encoded, numerical features were standardized, and feature importance was analyzed to understand the contribution of each variable towards employee attrition prediction.

### 6.1 Label Encoding

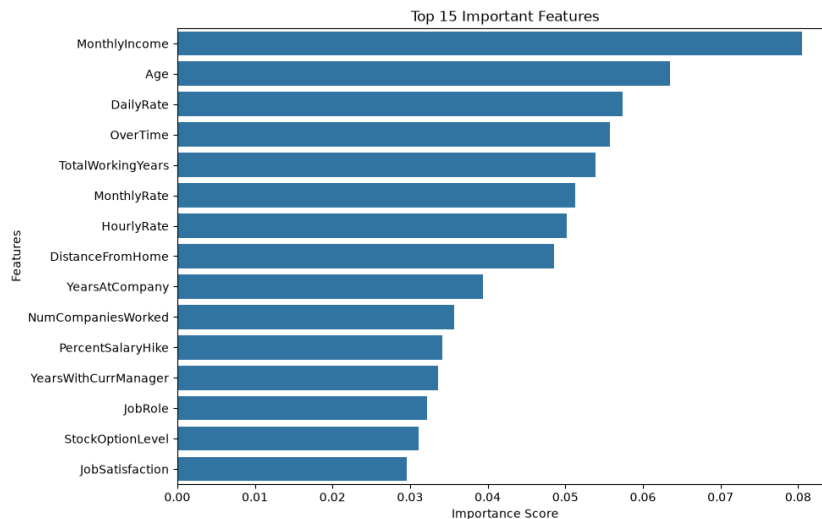
Several categorical attributes in the dataset, including **BusinessTravel**, **Department**, **EducationField**, **Gender**, **JobRole**, **MaritalStatus**, **Over18**, and **OverTime**, were converted into numerical values using **LabelEncoder** from the Scikit-learn library. Separate encoders were fitted for each feature to preserve the uniqueness of category mappings. These encoders were saved as **encoders.pkl**, allowing the same transformation to be applied during prediction in the deployed application.

### 6.2 StandardScaler

After encoding the categorical variables, feature scaling was performed using **StandardScaler**. Standardization transformed all numerical features to have a mean of zero and a standard deviation of one, ensuring that every feature contributed fairly during model training. The fitted scaler was stored as **scaler.pkl**, while the feature order was saved in **features.pkl** to maintain consistency between training and deployment.

### 6.3 Feature Importance

Feature importance analysis was performed using a Random Forest model to identify the variables that contributed most to employee attrition prediction. The analysis showed that features such as **Monthly Income**, **Age**, **Daily Rate**, **OverTime**, and **Total Working Years** had the greatest influence on the model's predictions. This step helped interpret the dataset and understand which employee characteristics were most significant.



*Figure 6.1 - Top 15 Important Features (Random Forest)*

## 6.4 Business Interpretation

The feature importance analysis indicates that employee attrition is influenced by a combination of financial, professional, and personal factors. Variables such as monthly income, work experience, overtime, and age play a significant role in predicting whether an employee is likely to leave the organization. These insights suggest that employee retention cannot be explained by a single factor but rather by the combined effect of multiple workplace conditions.

## 6.5 Business Recommendations

Based on the feature engineering and analysis, the following recommendations can help improve employee retention:

- Reduce excessive overtime by maintaining a balanced workload.
- Review salary structures to ensure fair and competitive compensation.
- Provide career development opportunities for experienced employees.
- Improve employee engagement and job satisfaction through regular feedback.
- Monitor high-risk employees using predictive analytics to take proactive retention measures.

## 6.6 Chapter Summary

This chapter described the feature engineering techniques applied to prepare the dataset for machine learning. Label encoding and feature scaling ensured that the data was suitable for model training, while feature importance analysis highlighted the most influential factors affecting employee attrition. These processed features formed the foundation for building and evaluating the predictive models discussed in the next chapter.

## CHAPTER 7: MODEL BUILDING

### 7.1 Logistic Regression

Logistic Regression is a linear classification algorithm that models the probability of an employee belonging to the “Yes” (attrition) class using a sigmoid function applied to a weighted sum of the input features. It is simple, fast to train, and its coefficients can be interpreted directly, which makes it a natural starting point for a binary classification problem such as this one.

### 7.2 Decision Tree

A Decision Tree Classifier splits the data recursively based on feature values that best separate the two classes, forming a tree-like structure of decisions. It is easy to interpret visually but can overfit the training data if not constrained.

### 7.3 Random Forest

Random Forest is an ensemble of many decision trees, each trained on a random subset of the data and features, with the final prediction decided by majority vote. It generally reduces overfitting compared to a single decision tree and was also used separately in this project to compute feature importance.

### 7.4 K-Nearest Neighbours

The K-Nearest Neighbours algorithm classifies a new employee record by looking at the labels of the most similar records in the training data, based on distance in the scaled feature space, and assigning the majority label among them.

### 7.5 Support Vector Machine

The Support Vector Machine algorithm finds a decision boundary that best separates the two classes with the maximum possible margin. In this project it was configured with `probability=True` so that it could also output class probabilities, not just hard predictions.

### 7.6 Why Logistic Regression Was Selected

Among all the evaluated machine learning algorithms, Logistic Regression achieved the highest classification accuracy of **87.41%**. In addition to its strong predictive performance, the

model is computationally efficient, easy to interpret, and well-suited for binary classification problems such as employee attrition prediction.

Another advantage of Logistic Regression is its ability to generate prediction probabilities, which were later used in the Streamlit application to display the likelihood of an employee leaving the organization. Considering its accuracy, simplicity, and deployment efficiency, Logistic Regression was selected as the final model for the Employee Attrition Prediction System.

| <b>Model</b>           | <b>Accuracy (%)</b> |
|------------------------|---------------------|
| Logistic Regression    | <b>87.41</b>        |
| Support Vector Machine | 85.37               |
| K-Nearest Neighbours   | 85.03               |
| Random Forest          | 83.67               |
| Decision Tree          | 78.23               |

## **7.7 Chapter Summary**

This chapter presented the implementation and comparison of five machine learning algorithms for predicting employee attrition. Each model was evaluated using the same dataset to identify the most suitable classifier. Based on the comparative analysis, Logistic Regression achieved the highest accuracy and was selected as the final model for deployment. The next chapter discusses the evaluation of the selected model using different performance metrics and prediction results.

## CHAPTER 8: MODEL EVALUATION

### 8.1 Accuracy

Accuracy measures the percentage of correctly classified employee records out of the total predictions made by the model. Among all the trained algorithms, Logistic Regression achieved the highest testing accuracy.

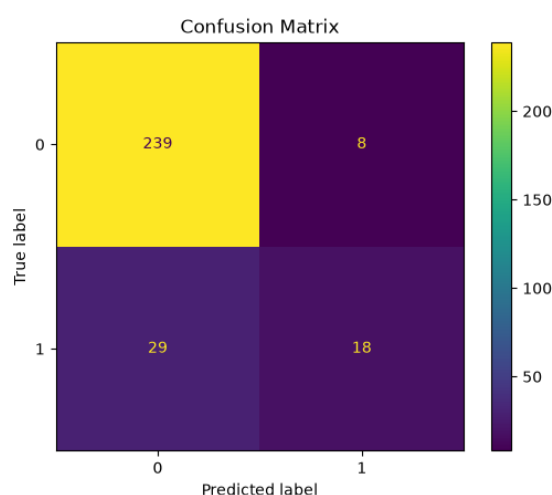
**Final Model Accuracy: 87.41%**

This result indicates that the model correctly predicts employee attrition for the majority of employees and demonstrates strong overall performance on the IBM HR Analytics dataset.

### 8.2 Confusion Matrix

The confusion matrix provides a detailed view of the model's prediction performance by comparing actual and predicted class labels. It shows the number of correctly classified employees along with the misclassified cases.

The matrix indicates that the Logistic Regression model successfully classified most employees while maintaining a relatively low number of incorrect predictions. This demonstrates the model's ability to distinguish employees who are likely to leave from those who are expected to remain with the organization.



*Figure 8.1 - Confusion Matrix - Logistic Regression*

### 8.3 Classification Report

The classification report evaluates the model using **Precision**, **Recall**, **F1-Score**, and **Support** for each class.

- **Precision** measures how many predicted attrition cases were actually correct.
- **Recall** measures how effectively the model identified employees who left the organization.
- **F1-Score** provides a balanced measure by combining precision and recall.

Overall, the classification report confirms that the Logistic Regression model achieved balanced performance across both classes and produced reliable predictions.

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| 0            | 0.89      | 0.97   | 0.93     | 247     |
| 1            | 0.69      | 0.38   | 0.49     | 47      |
| accuracy     |           |        | 0.87     | 294     |
| macro avg    | 0.79      | 0.68   | 0.71     | 294     |
| weighted avg | 0.86      | 0.87   | 0.86     | 294     |

Table 8.1: Classification Report

### 8.4 Probability Prediction

One of the advantages of Logistic Regression is its ability to predict probabilities instead of only class labels. For every employee, the model calculates the probability of **staying** and **leaving** the organization.

For example, during testing, the model generated predictions such as:

- **Probability of Staying:** 28.91%
- **Probability of Leaving:** 71.09%

These probability scores were also integrated into the Streamlit application, allowing users to understand the confidence level behind each prediction.

## 8.5 Model Comparison

Five machine learning algorithms were trained and compared during this project. Logistic Regression achieved the highest prediction accuracy and outperformed the remaining classifiers.

| Model                  | Accuracy (%) |
|------------------------|--------------|
| Logistic Regression    | 87.41        |
| Support Vector Machine | 85.37        |
| K-Nearest Neighbours   | 85.03        |
| Random Forest          | 83.67        |
| Decision Tree          | 78.23        |

The comparison demonstrates that Logistic Regression provides the best balance between prediction accuracy, computational efficiency, and ease of deployment.

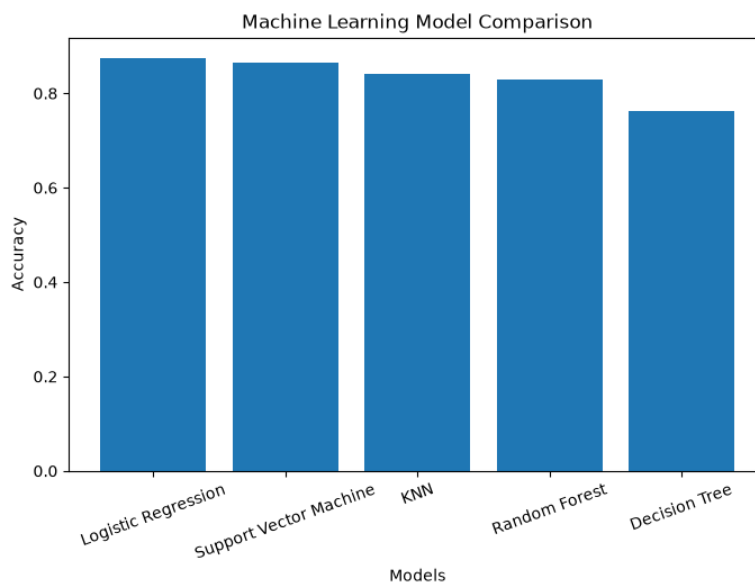
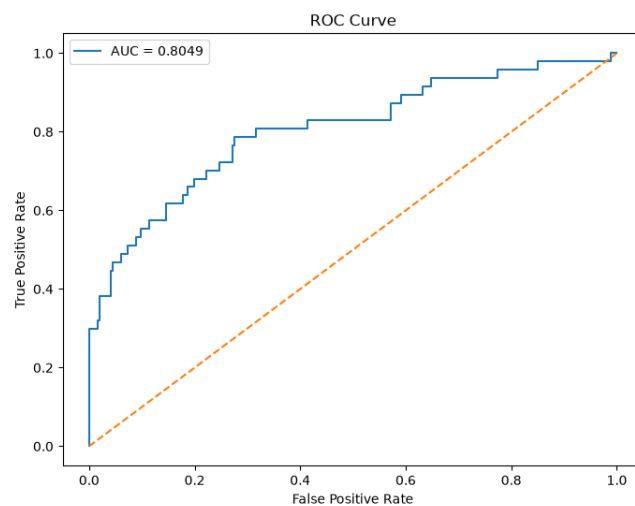


Figure 8.2: Accuracy Comparison Chart

## 8.6 Final Performance

Based on all evaluation metrics, Logistic Regression was selected as the final model for deployment. The model consistently delivered accurate predictions while maintaining good interpretability and fast execution time.

The trained model was successfully integrated into a Streamlit web application, where users can enter employee details and receive real-time attrition predictions along with prediction probabilities.



*Figure 8.3 - ROC Curve - Logistic Regression*

## 8.7 Advantages

The developed Employee Attrition Prediction System offers several advantages:

- High prediction accuracy of **87.41%**.
- Fast and efficient prediction using Logistic Regression.
- User-friendly Streamlit interface for real-time predictions.
- Probability-based prediction for better decision-making.
- Easy deployment and maintenance through GitHub and Streamlit Community Cloud.
- Supports HR professionals in identifying employees at risk of leaving.

## 8.8 Limitations

Although the developed system performs well, certain limitations remain:

- The model was trained using only the IBM HR Analytics dataset.
- Real-world organizational policies and external economic factors were not included.
- Employee behaviour may change over time, requiring periodic model retraining.
- The model should be considered a decision-support tool rather than a replacement for human judgment.

## 8.9 Chapter Summary

This chapter evaluated the performance of the developed Employee Attrition Prediction System using multiple evaluation metrics. Logistic Regression achieved the highest prediction accuracy of **87.41%** and demonstrated reliable classification performance through the confusion matrix, classification report, and probability-based predictions. The encouraging evaluation results confirmed the suitability of the model for deployment and practical HR decision support.

## CHAPTER 9: STREAMLIT APPLICATION

## 9.1 Architecture

The application follows a simple machine learning deployment architecture. User inputs are collected through the Streamlit interface, preprocessed using the saved encoders and scaler, and then passed to the trained Logistic Regression model. Finally, the prediction result and probability scores are displayed to the user.

## 9.2 User Interface

The Streamlit application provides a clean and user-friendly interface for entering employee information. Input fields are created using dropdown menus, sliders, and number input boxes, making the application easy to use without requiring technical knowledge.

The interface displays:

- Employee information form
- Prediction button
- Attrition prediction result
- Probability of staying and leaving
- Risk indicator for better interpretation

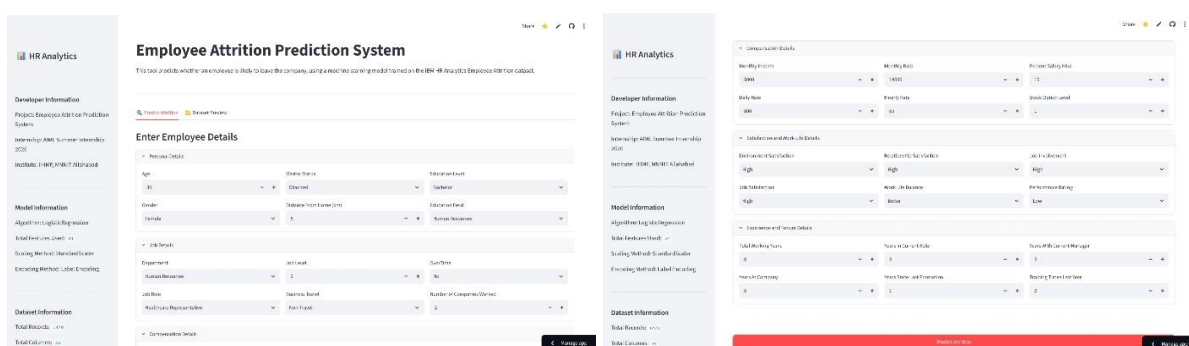


Figure 9.1 - Streamlit Dashboard - Home Page

### 9.3 Prediction Flow

Once the user submits the employee details, the application first applies the same preprocessing steps used during model training. Categorical variables are encoded, numerical values are

standardized, and the feature order is maintained before passing the data to the trained Logistic Regression model.

The model predicts whether the employee is likely to stay or leave the organization and also calculates the prediction probability. The final result is displayed instantly through the web interface.

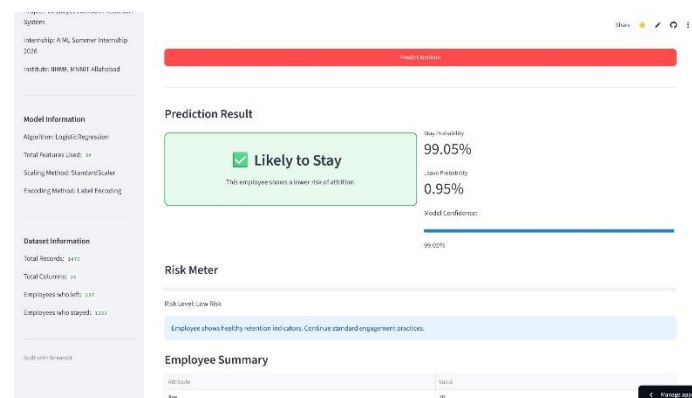


Figure 9.2 - Streamlit Dashboard - Prediction Result

## 9.4 Deployment Process

The completed application was deployed using **Streamlit Community Cloud**. The project files, trained model, encoders, scaler, feature list, and application code were uploaded to GitHub. The GitHub repository was then connected to Streamlit Community Cloud, allowing the application to be deployed online without requiring any local installation.

The deployment process included:

- Uploading project files to GitHub
- Creating a requirements.txt file
- Connecting the GitHub repository with Streamlit Community Cloud
- Deploying the application using app.py as the entry point

## 9.5 GitHub Integration

GitHub was used for version control and project management throughout the development process. All notebooks, datasets, trained models, deployment files, and application source code were maintained in a single repository, ensuring organized project documentation and easy collaboration.

The GitHub repository also serves as a permanent source for future updates and deployment.

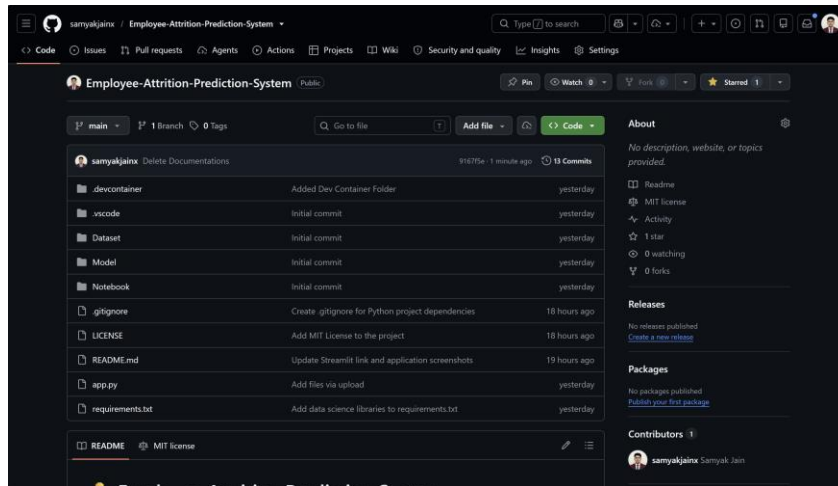


Figure 9.3: GitHub Repository Screenshot

## 9.6 Application Features

The developed Streamlit application provides several useful features:

- Interactive and user-friendly interface.
- Real-time employee attrition prediction.
- Prediction probability for both staying and leaving.
- Automatic preprocessing using saved encoders and scaler.
- Fast prediction using the trained Logistic Regression model.
- Cloud deployment for easy online accessibility.
- Integrated GitHub repository for version control and maintenance.

## 9.7 Chapter Summary

This chapter presented the development and deployment of the Streamlit-based Employee Attrition Prediction System. The application successfully integrates the trained Logistic Regression model with an interactive web interface, enabling users to obtain real-time predictions with minimal effort. Deployment through Streamlit Community Cloud and GitHub makes the system easily accessible, maintainable, and suitable for practical HR analytics applications.

## CHAPTER 10: RESULTS AND DISCUSSION

### 10.1 Model Performance

The performance of all trained models was compared using classification accuracy. Logistic Regression achieved the highest accuracy of **87.41%**, outperforming Decision Tree, Random Forest, K-Nearest Neighbours, and Support Vector Machine. The model also produced consistent results through the confusion matrix and classification report, making it suitable for employee attrition prediction.

### 10.2 Business Insights

The analysis revealed that employee attrition is influenced by several factors, including **Monthly Income, OverTime, Age, Total Working Years, Job Satisfaction, and Work-Life Balance**. Employees working overtime or receiving lower salaries showed a higher likelihood of leaving the organization. These insights can help HR departments identify potential risks and implement proactive employee retention strategies.

### 10.3 Practical Applications

The developed system can assist Human Resource departments in identifying employees who may be at risk of leaving the organization. By predicting attrition in advance, organizations can improve workforce planning, reduce recruitment costs, and take preventive measures to increase employee satisfaction and retention. The deployed Streamlit application makes these predictions easily accessible through a simple web interface.

### 10.4 Overall Findings

The project successfully demonstrated the use of machine learning in Human Resource Analytics. The combination of effective data preprocessing, feature engineering, model evaluation, and web deployment resulted in a practical and reliable Employee Attrition Prediction System. The project highlights the potential of predictive analytics in supporting data-driven HR decision-making.

## CHAPTER 11: CONCLUSION

This project presented the design and development of an **Employee Attrition Prediction System** using machine learning techniques. The IBM HR Analytics dataset was preprocessed, analyzed, and used to train multiple classification models. After comparing their performance, **Logistic Regression** achieved the highest accuracy of **87.41%** and was selected as the final model.

The trained model was successfully integrated into a Streamlit web application, allowing users to predict employee attrition through an interactive interface. The application provides real-time predictions along with probability scores, making the results more meaningful and easy to interpret.

Overall, this project enhanced my understanding of data preprocessing, exploratory data analysis, feature engineering, machine learning model development, evaluation, and deployment. It also demonstrated how machine learning can be applied to solve practical business problems in Human Resource Analytics.

## CHAPTER 12: FUTURE SCOPE

### 12.1 Explainable AI (SHAP)

Future versions of the project can integrate Explainable AI techniques such as **SHAP (SHapley Additive Explanations)** to provide detailed explanations for each prediction. This would help HR professionals understand the factors influencing employee attrition and improve confidence in the model's recommendations.

### 12.2 Deep Learning

Deep learning models such as Artificial Neural Networks (ANNs) can be explored to capture more complex relationships within employee data. These models may further improve prediction performance when larger and more diverse datasets are available.

### 12.3 HR Dashboard

An interactive HR dashboard can be developed to visualize employee statistics, attrition trends, department-wise analysis, and prediction summaries. Such a dashboard would support management in monitoring workforce performance and making informed decisions.

### 12.4 Database Integration

The current application uses manually entered employee data. Future versions can be integrated with databases such as MySQL or PostgreSQL, enabling automatic retrieval and storage of employee records for large-scale organizational use.

### 12.5 Authentication

User authentication and role-based access control can be implemented to improve security. This would ensure that only authorized HR personnel can access employee information and prediction results.

### 12.6 Cloud Improvements

The application can be deployed on advanced cloud platforms such as Microsoft Azure, AWS, or Google Cloud to improve scalability, reliability, and availability for enterprise-level deployment.

## **12.7 Real-time Prediction**

Future versions can support real-time employee monitoring by integrating with HR Management Systems (HRMS). This would enable continuous analysis of employee data and provide instant attrition predictions whenever employee information is updated.

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